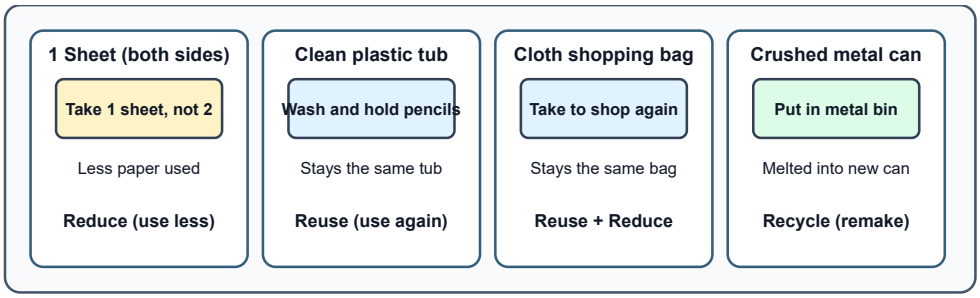
Science · BUILD · Autumn 2 · Week 1 · SCI_BUILD_A2_W01_P1

Knowledge organiser



Knowledge organiser model comparing reduce, reuse, and recycle across paper, a plastic tub, a cloth bag, and a metal can.

Learning goal: I can describe how to care for a place by using less, using safe items again, and sorting waste for recycling.

Tier 2 and Tier 3 vocabulary

Tier	Word	Meaning	Example in this lesson
Tier 2	reduce	To use less of a thing.	I write on the blank back so I need less paper.
Tier 3	reuse	To use a clean, safe thing again.	I fill a clean tub with pencils to use it again.
Tier 3	recycle	To make new things from old material.	A clean metal can goes in the bin to make a new thing.
Tier 3	material	What a thing is made of.	This drink can is made of metal.

Core science ideas: Reduce means using less material at the start. Reuse means using the same object again without remaking it. Recycle means processing waste material at a factory to make a new product.

Keep claims careful: Refilling a bottle or using a tub for pencils is reuse, not recycle. Recycling comes after reduce and reuse because factories still need energy and clean sorting.

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Teacher answers and topic guidance

Accept any scientifically equivalent wording in spoken, signed, pointed, drawn, or written form.

Starter answer

Option A (Reuse). Washing a plastic tub and keeping pencils in it uses the same tub again without melting or remaking the plastic.

Arrival · Supported

1. Tray B shows a clean tub holding pencils.
2. Animals get food from what they eat.
3. Muscles pull on bones to help us move.
4. Yes. We can write on the blank back to use less paper.

Arrival · Independent

1. Tray A shows a paper sheet with a blank back.
2. Animals get food from what they eat, while green plants make their own food using light.
3. The skeleton holds the body up and muscles pull on bones so we can move.
4. Using the clean tub again holds pencils without needing to buy or make a new tub.

Arrival · Aspire

1. Tray A uses the blank back so we take no new paper, and Tray B reuses a clean tub so we need no new pot.
2. Animals cannot make their own food, so they must eat to get energy and materials for their bodies.
3. A muscle can contract and pull a bone across a joint, so another muscle pulls it back.
4. We must check that the tub is clean and has no sharp or broken edges.

You do · Supported mark scheme

1. The paper with a blank back matches Reduce (use less). *(Up to 2 marks)*
2. We can reuse the clean tub to hold pencils, or reuse the cloth bag to carry books. *(Up to 2 marks)*
3. I can use the tub again because it is clean and safe. *(Up to 2 marks)*

You do · Independent mark scheme

1. Reduce: sheet with blank back. Reuse: clean tub and cloth bag. Recycle: empty metal can (after checking local bin rules). *(Up to 3 marks)*
2. Using the cloth bag again is reuse. It saves new material because we do not need to make a fresh bag. *(Up to 3 marks)*
3. First check that the item is clean and has no sharp edges. Second check the local bin label to see if that material is accepted. *(Up to 2 marks)*

You do · Choice mark scheme

1. 1. Reduce (use less). 2. Reuse (use again). *(Up to 2 marks)*
2. 3. Metal. 4. Local bin label. *(Up to 2 marks)*
3. Use a clean cloth bag again because it is clean and needs no new material. *(Up to 2 marks)*

You do · Aspire challenge mark scheme

1. Reusing keeps the same clean tub and uses it again without changing it. Recycling breaks the old plastic down at a factory to make a new item. *(Up to 2 marks)*
2. Using less and reusing a safe item stop waste right away without needing lorries and factory energy to melt and remake the material. *(Up to 2 marks)*

Exit ticket · Supported

1. That is Reduce (use less).
2. Yes. That is reuse.
3. We check the local bin label.

Exit ticket · Independent

1. Reduce: write on the blank back of a sheet. Reuse: fill a clean tub with pencils or use a clean cloth bag again.
2. Material means what a thing is made of, such as metal in a can or paper in a sheet.
3. Different places accept different materials, and the wrong item can spoil the recycling load.

Exit ticket · Aspire

1. Reuse uses the same clean item again as it is, while recycle processes old material into a new product.
2. Refilling reduces the number of new bottles made and avoids the energy needed to collect and reprocess five bottles.
3. An unchecked waste item could be dirty or have sharp edges that could hurt someone.

Topic guidance and common misconceptions

- **Reuse vs recycle:** Pupils often call any positive environmental action 'recycling'. Emphasise the object test: if the bottle or tub stays the same object, it is **reuse**; if a factory melts or pulps the material to make a new thing, it is **recycle**.
- **Why reduce and reuse come first:** Recycling still uses collection trucks, heat, water, and electricity. Reducing and reusing save both raw materials and factory processing energy.
- **Clean and dry sorting:** Food or liquid left inside containers can spoil a whole load of paper or card in a recycling bin.

Made by Matt